

CLINISCAN: LUNG ABNORMALITY DETECTION ON CHEST X-RAYS USING AI

1. Introduction

Medical imaging plays a critical role in the diagnosis of pulmonary diseases. Chest X-rays (CXR) are one of the most commonly used imaging modalities due to their low cost, speed, and wide availability. Manual interpretation of X-ray images is time-consuming and highly dependent on the expertise of radiologists.

To address these challenges, this project utilizes a large-scale, annotated chest X-ray dataset to develop an AI-based lung abnormality detection system.

2. Dataset Source

The dataset used in this project is **VinBigData Chest X-ray Abnormalities Detection**, obtained from Kaggle.

3. Dataset Overview

- **Dataset Name:** VinBigData Chest X-ray Abnormalities Detection
- **Total Images:** 15,000 chest X-ray images
- **Image Format:** DICOM (.dicom)
- **Image Modality:** Chest X-ray (CXR)
- **Annotation Format:** CSV file with bounding box coordinates
- **Annotation Type:** Object detection (localization) + classification

4. Image Characteristics

- Images are stored in DICOM (Digital Imaging and Communications in Medicine) format.
- High-resolution grayscale medical images.
- Pixel intensity values vary depending on acquisition devices.
- Images represent frontal chest X-ray views.

5. Annotations

Each image is annotated by expert radiologists. The annotations include:

- `image_id` - unique image identifier
- `class_name` - the name of the class of detected object (or "No finding")
- `class_id` - the ID of the class of detected object
- `rad_id` - the ID of the radiologist that made the observation
- `x_min` - minimum X coordinate of the object's bounding box
- `y_min` - minimum Y coordinate of the object's bounding box
- `x_max` - maximum X coordinate of the object's bounding box
- `y_max` - maximum Y coordinate of the object's bounding box

Abnormality Classes (Examples)

0 - Aortic enlargement

1 - Atelectasis

2 - Calcification

3 - Cardiomegaly

4 - Consolidation

5 - ILD

6 - Infiltration

7 - Lung Opacity

8 - Nodule/Mass

9 - Other lesion

10 - Pleural effusion

11 - Pleural thickening

12 - Pneumothorax

13 - Pulmonary fibrosis

14 - No finding

Images labeled as “**No finding**” do not contain bounding box annotations.

6. Clinical Relevance

This dataset is clinically significant because:

- It contains real-world hospital data
- Annotations are provided by multiple radiologists
- It enables detection of life-threatening lung and cardiac conditions
- Suitable for developing AI-assisted diagnostic systems

In the CliniScan project:

- A subset of images is initially used for experimentation
- DICOM images are converted into PNG format
- Annotations are transformed **into YOLO format**
- Data is prepared for deep learning-based detection models

8. Ethical Considerations

- Dataset is de-identified
- No patient-specific personal information is present
- Usage complies with open-data and research guidelines